(1E)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

hod.ae@rvce.edu.in
www.rvce.edu.in
Tel: 080-68188240

Department of Aerospace Engineering

DEPARTMENT OF
AEROSPACE ENGINEERING

Date	18th June 2024	Maximum Marks	50
Course Code	AS343AI	Duration	90 Min
Sem	IV Semester		

AEROSPACE PROPULSION

PART B

1	Identify the various factors which cause the Brayton cycle to deviate from its ideal conditions.	10	L1	CO1
2	Highlight the various scenarios where a turbojet engine with basic thrust cannot function. What are the technical remedies for this? Explain anyone technique with appropriate diagrams.	10	L2	CO2
3	Describe the working of a turbofan engine highlighting all the components. List the characteristics of the engine.	10	L2	CO2
4	Consider a control volume of a propulsive device with Area ai and aj at the entry and exit. Derive an expression to find the total thrust for this propulsive device. Additionally, explain the following terms associated with the engine giving its mathematical expressions: • Specific thrust • Effective speed ratio • Specific impulse	10	L3	CO3
5	3) The diameter of the propeller of an aircraft is 2.5m. It flies at a speed of 500kmph at an altitude of 8000m. For a flight to jet speed ratio of 0.75, determine • Flow rate of air through the propeller • Thrust • Specific thrust • Specific impulse • Thrust power At h=8000m, take ρ=0.525kg/m^3	10	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	10	10	10	20	20	--	--	--

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

hod.ae@rvce.edu.in
www.rvce.edu.in
Tel: 080-68188240

Department of Aerospace Engineering

22AS057

DEPARTMENT OF
AEROSPACE ENGINEERING

Date	23rd July 2024	Maximum Marks	50
		Duration	90 Min
Course Code	AS343AI	Offline Mode	
Sem	IV Semester		

AEROSPACE PROPULSION CIE 2

PART B				
1	Explain how shockwaves are used for achieving supersonic and subsonic diffusion in supersonic inlets of aircraft. Additionally, describe the working of a normal shock inlet based on the position of shockwave.	10	L1	CO2
2	For a given mass flow rate and the expansion ratio, which is the efficient turbine, Impulse or reaction? With a neat sketch, show the variation of pressure and velocity through a reaction turbine and also explain its working.	10	L2	CO3
3	Describe the working of a centrifugal compressor employed for achieving compression in a jet engine highlighting its advantages and disadvantages. Illustrate all the necessary diagrams to support the same.	10	L1	CO1
4	A simple turbojet engine operates with a maximum turbine inlet temperature of 1200K , a pressure ratio of 4.25:1 and a mass flow rate of 25kg/s under design conditions, the following efficiencies can be assumed. • Isentropic efficiency of the compressor: 87% • Isentropic efficiency of turbine: 91.5% • Propelling nozzle efficiency: 96.5% • Transmission efficiency: 98.5% • Combustion chamber pressure loss: 0.21Bar Assume C_{Pa}=1.005kJ/kgK and γ_a=1.4, C_{Pg}=1.147kJ/kgK and γ_g=1.3. Calculate the total design thrust.	10	L3	CO4
5	Justify, why the number of compressor stages always more than the turbine stages. Also, explain the reason for the turbine efficiency being always higher than the compressor efficiency. Besides, explain the pressure and velocity variation through an axial flow compressor with detailed explanation.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	11	22	11	16	21s	22	17	--	--	--	--

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

hod.ae@rvce.edu.in
www.rvce.edu.in
Tel: 080-68188240

Department of Aerospace Engineering

DEPARTMENT OF
AEROSPACE ENGINEERING

Date	26th August 2024	Maximum Marks	50
Course Code	AS343AI	Duration	90 Min
Sem	IV Semester		

AEROSPACE PROPULSION CIE-3

PART B

1	Explain with a neat sketch, the working of a Pyrogen Igniter highlighting its importance in rocket combustion process.	10	L1	CO1
2	It is a well-known fact that the thrust requirement varies with the service altitude of the rocket. In case of a solid propellant rocket, explain how thrust is controlled and varied in accordance with the operating altitude. Also, highlighting the same with illustrations and examples would be beneficial.	10	L2	CO2
3	Illustrate on a neat diagram and explain the working of turbo-pump propellant feed systems for large scale rocket engine applications.	10	L1	CO2
4	Imagine a situation wherein a 4 stage solid rocket motor is being launched into the geo synchronous orbit with a payload. However, the motor loses its stability in the first phase and goes out of control. To prevent danger of falling back to earth, what would be the possible techniques that can be employed for the thrust extinction of the motor?	10	L3	CO3
5	In a turbojet unit with forward facing ram intake, the jet velocity relative to the propelling nozzle at exit is twice the flight velocity. Determine the rate of fuel consumption in kg/s, when developing a thrust of 25000 N under the following conditions. • Ambient pressure and temperature : 0.7 bar; 1°C • Compression total head pressure ratio : 5:1 • Flight speed : 800 km/h • CV of fuel : 42000 kJ/kg • Ram efficiency : 100% • Isentropic efficiency of compressor : 85% • Isentropic efficiency of turbine : 90% • Isentropic efficiency of nozzle : 95% • Combustion efficiency : 98% • Turbine pressure ratio : 2.23 Assume the mass flow of fuel is small compared with the mass flow of air and that the working fluid throughout has the properties of air at low temperature. Neglect the extraneous pressure drop. Assume $Cp_g=Cp_a=1.005$ kJ/kg K.	10	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	10	10	20	10	10	10	--	--